

Nassau Candy Distributor – Product Line Profitability & Margin Performance Analysis

Research Paper

EDA , Insights and Recommendations

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1.Executive Summary

This study evaluates product-level profitability and margin performance for Nassau Candy Distributor using sales, cost, units, gross profit, product, and division data. The analysis focuses on identifying profitable products, low-margin products, division-level performance, profit and revenue concentration, and cost-related margin risks. The dataset records total sales of \$141,783.63, gross profit of \$93,442.80, and 38,654 units, with an overall average gross margin of 67%. The analysis also shows that 5 products, representing 33.33% of the product portfolio, account for at least 80% of total profit and 80% of total revenue. Margin volatility over the observed monthly periods is approximately 0.32 percentage points. These findings provide a data-driven basis for monitoring high-contribution products, reviewing low-margin products, and supporting pricing and cost-management decisions.

2.Introduction

Profitability analysis is essential for understanding whether sales volume is translating into sustainable financial performance. For a distributor such as Nassau Candy, products may generate high sales while contributing relatively little profit because of high costs or weak margins. Therefore, evaluating sales alone is not sufficient for effective product and portfolio decisions.

This study analyzes product-level and division-level profitability using sales, cost, units, gross profit, and margin information. The analysis examines product profitability, margin performance, revenue and profit contribution, cost-to-sales relationships, profit concentration, and margin variability over time.

The objective is to identify products and divisions that contribute strongly to financial performance, detect products with potential margin risk, and provide evidence-based recommendations for pricing, cost management, and product portfolio decisions.

3.Dataset Description

3.1 Dataset Description

The analysis uses the Nassau Candy Distributor dataset, which contains order-level information about products, customers, sales, costs, units, and gross profit. Important fields include Order Date, Ship Date, Division, Region, Product ID, Product Name, Sales, Units, Gross Profit, and Cost. These fields provide the basis for evaluating product profitability, margin performance, division-level performance, and cost structure.

3.2 Data Preparation

The dataset was reviewed to validate sales, cost, units, and gross profit values. Invalid or unsuitable records were considered during the analysis, and product and division information was standardized for consistent aggregation. Gross margin was calculated as Gross Profit divided by Sales, while profit per unit was calculated as Gross Profit divided by Units.

3.3 Analytical Approach

The analysis combines product-level, division-level, concentration, and cost-structure analysis. PivotTables and PivotCharts were used to summarize the data and enable interactive filtering by division, product, and order date. Product profitability was examined using gross profit, gross margin, profit per unit, revenue contribution, and profit contribution. Cost-versus-sales and cost-versus-margin scatter analyses were used to identify potential cost and margin risks. Pareto analysis was used to determine the proportion of products contributing to 80% of total revenue and profit. Monthly margin values were also examined to measure margin volatility over time.

4.Exploratory Data Analysis

4.1 Overall Performance

The dataset shows total sales of \$141,783.63, gross profit of \$93,442.80, and 38,654 units sold. The overall average gross margin is 67%, indicating that the business generates a substantial gross profit relative to sales. Margin volatility across the observed monthly periods is approximately 0.32 percentage points, suggesting relatively limited variation in the average monthly margin. These overall measures provide a baseline for evaluating product-level and division-level profitability.

4.2 Product-Level Profitability

Product-level analysis shows differences in sales, gross profit, and margin across the product portfolio. The product profitability analysis indicates that products can have different financial characteristics even when their sales levels are similar. The classification analysis identifies high-profit/high-margin products, high-sales/low-margin products, and low-sales/low-profit products. High-sales/low-margin products require particular attention because strong sales volume does not necessarily translate into proportionally strong profitability. Low-sales/low-profit products may also require portfolio review based on their contribution and margin performance.

4.3 Division-Level Performance

Division-level analysis shows differences in revenue, gross profit, and average margin across the three divisions. The Chocolate division generates the highest revenue and gross profit among the divisions, while the Other division records lower performance on these measures. The Sugar division also contributes substantially to overall revenue and profit. Average margin differs across divisions, indicating that revenue scale and margin performance should be considered together when evaluating division-level financial performance.

4.4 Cost and Margin Diagnostics

Cost and margin diagnostics were used to examine the relationship between product costs, sales, and profitability. The cost-versus-sales analysis shows how product costs vary with sales levels, while the cost-versus-margin analysis helps identify products with relatively high costs and lower margins. The margin risk analysis identifies five products with average margins at or below 50%: Kazooles, Fun Dip, SweetTARTS, Nerds, and Lickable Wallpaper. These products should be reviewed for pricing, cost-reduction, or other margin-improvement opportunities.

4.5 Profit and Revenue Concentration

Pareto analysis was used to examine the concentration of profit and revenue across products. The analysis shows that 5 products, representing 33.33% of the product portfolio, account for at

least 80% of total gross profit. The same 5 products, also representing 33.33% of the portfolio, account for at least 80% of total revenue. This indicates that a relatively small portion of the product portfolio contributes a large share of overall financial performance and should therefore be monitored for dependency risk.

4.6 Margin Volatility

Monthly average gross margin was examined to assess the stability of margin performance over time. The calculated margin volatility is approximately 0.32 percentage points across the observed monthly periods. The relatively small variation indicates that average monthly margins remained within a comparatively narrow range during the observation period. Monitoring margin trends over time can help identify changes in pricing, product mix, or cost conditions that may affect profitability.

5.Key Findings and Insights

5.1 Overall Profitability

The analysis indicates that Nassau Candy Distributor generated \$141,783.63 in total sales and \$93,442.80 in gross profit from 38,654 units. The overall average gross margin was 67%. These results indicate that the product portfolio generated substantial gross profit relative to sales, while the product-level analysis shows that profitability is not distributed evenly across all products.

5.2 Product-Level Insights

The product-level analysis shows that profitability varies considerably across the product portfolio. High-profit/high-margin products contribute strongly to both sales and profitability and should be monitored for continued availability and performance. High-sales/low-margin products generate substantial sales but contribute relatively less margin, making them important candidates for pricing or cost review. Low-sales/low-profit products contribute less to overall financial performance and may require portfolio evaluation. The analysis also identified five products with average margins at or below 50%: Kazooles, Fun Dip, SweetTARTS, Nerds, and Lickable Wallpaper. These products represent potential margin-risk areas and should be reviewed for pricing and cost-management opportunities.

5.3 Division-Level Insights

Division-level analysis shows differences in revenue, gross profit, and average margin across the three divisions. The Chocolate division contributes strongly to overall revenue and gross profit, while the Sugar and Other divisions also make significant contributions. Comparing revenue and profit together provides a clearer view of division performance than considering sales alone. Differences in average margin also indicate that some divisions may require greater attention to cost structure and pricing strategies. Therefore, division-level performance should be evaluated using revenue, gross profit, and margin together.

5.4 Cost and Margin Insights

Cost and margin diagnostics show that product costs have an important relationship with sales and profitability. Products with relatively high costs and lower margins can reduce overall profitability even when their sales are significant. The cost-versus-sales analysis helps identify products with higher cost levels, while the cost-versus-margin analysis highlights products with potential margin pressure. The five products identified with margins at or below 50% should therefore be reviewed for pricing adjustments, cost reduction, or other margin-improvement actions.

5.5 Profit and Revenue Concentration Insights

The Pareto analysis shows that profit and revenue are concentrated among a relatively small number of products. Five products, representing 33.33% of the total product portfolio, account for at least 80% of total gross profit. The same five products, also representing 33.33% of the portfolio, account for at least 80% of total revenue. This concentration indicates that the business depends significantly on a smaller group of products for its financial performance. These products should therefore be monitored carefully to reduce potential dependency and availability risks.

5.6 Margin Volatility Insights

The monthly margin analysis shows that the average gross margin remained relatively stable during the observed period. The calculated margin volatility is approximately 0.32 percentage points. This indicates that the average monthly margin did not fluctuate substantially over the analysis period. However, continued monitoring of margin trends is important because changes in product mix, pricing, or costs may affect future profitability.

6. Business Recommendations

Based on the findings from the exploratory and profitability analysis, several recommendations can be made to improve product and portfolio performance. High-contribution products should be monitored closely because a relatively small group of products generates a large share of total revenue and profit. Products with high sales but relatively low margins should be reviewed for pricing and cost-reduction opportunities. Products with margins at or below 50% should receive specific attention to identify opportunities for improving profitability. Division-level performance should be monitored using revenue, gross profit, and average margin together rather than relying on sales alone. Regular monitoring of margin trends can also help identify changes in pricing, product mix, and cost structure.

6.1 Project High-Contribution Products

The analysis shows that a relatively small number of products contribute a large share of total revenue and gross profit. These high-contribution products should be monitored regularly for sales performance, margin changes, inventory availability, and cost increases. Maintaining their availability and profitability can help reduce the risk of disruption to overall business performance.

6.2 Review Low-Margin Products

Five products were identified with average margins at or below 50%: Kazooles, Fun Dip, SweetTARTS, Nerds, and Lickable Wallpaper. These products should be reviewed to understand whether low margins are caused by high costs, pricing levels, or other product-specific factors. Management can consider appropriate pricing adjustments, supplier cost negotiations, or other cost-reduction measures where applicable.

6.3 High-Sales/Low-Margin Products

High-sales/low-margin products require particular attention because strong sales volume does not always result in proportionally strong profitability. These products should be examined to determine whether their margins can be improved through pricing adjustments, supplier negotiations, cost reduction, or changes in the product mix. Regular monitoring can help ensure that high sales volumes contribute effectively to overall profit.

6.4 Product Portfolio Review

Products with low sales and low profit contribution should be evaluated as part of the overall product portfolio. Management should examine their demand, strategic importance, operating costs, and contribution before deciding whether to retain, improve, reposition, or discontinue them. Portfolio decisions should be based on profitability and business relevance rather than sales volume alone.

6.5 Division and Margin Monitoring

Division performance should be monitored using revenue, gross profit, and average margin together. Comparing these measures can reveal differences between sales performance and profitability across divisions. Regular monitoring of division-level margins can help identify changes in cost structure, pricing performance, and product mix, allowing management to respond to emerging profitability risks.

6.6 Monitor Profit Concentration Risk

The analysis shows that five products, representing 33.33% of the product portfolio, account for at least 80% of total gross profit. This concentration creates a dependency on a relatively small group of products. Management should monitor the sales, margins, costs, and availability of these products and consider opportunities to strengthen other profitable products. This can help reduce excessive dependence on a small number of products.

6.7 Continuous Margin Monitoring

Margin performance should be monitored regularly to identify changes in profitability over time. Although the observed margin volatility is approximately 0.32 percentage points, future changes in product mix, pricing, and costs may affect margins. A regular review of monthly margin trends can help management identify emerging risks and take corrective action when necessary.

7.Conclusion

This study evaluated the profitability and margin performance of Nassau Candy Distributor using product-level and division-level analysis. The analysis examined sales, costs, units, gross profit, margins, product contribution, division performance, cost relationships, profit and revenue concentration, and margin volatility. The results show that the business generated \$141,783.63 in sales and \$93,442.80 in gross profit, with an overall average margin of 67%. Five products, representing 33.33% of the product portfolio, account for at least 80% of both total profit and total revenue. The analysis also identified five products with margins at or below 50% that require further review. Overall, the findings demonstrate the importance of combining sales, cost, margin, and contribution measures when evaluating product profitability. The recommendations provide a basis for improving pricing decisions, managing costs, monitoring high-contribution products, and reducing profitability risks.